

Guided tutorials

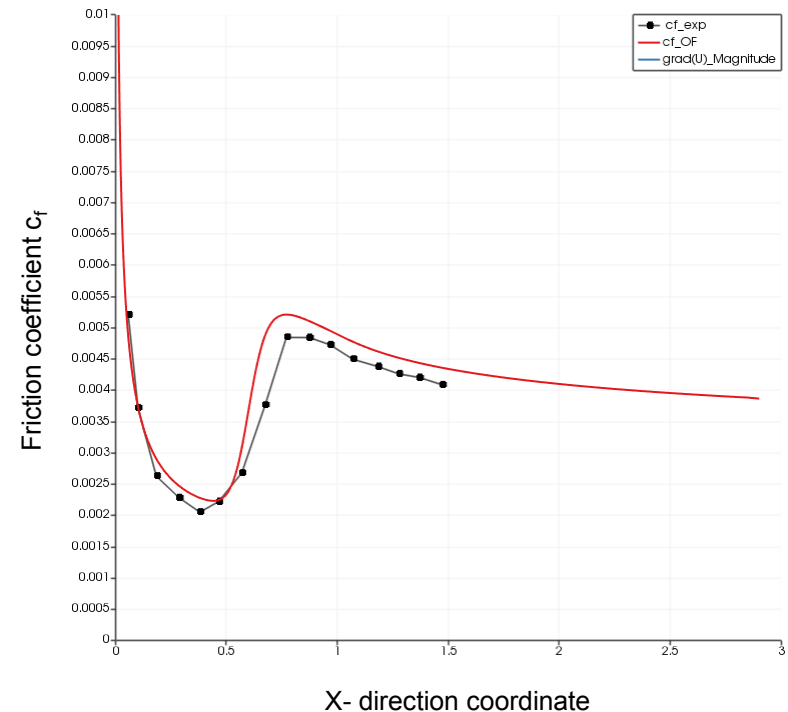
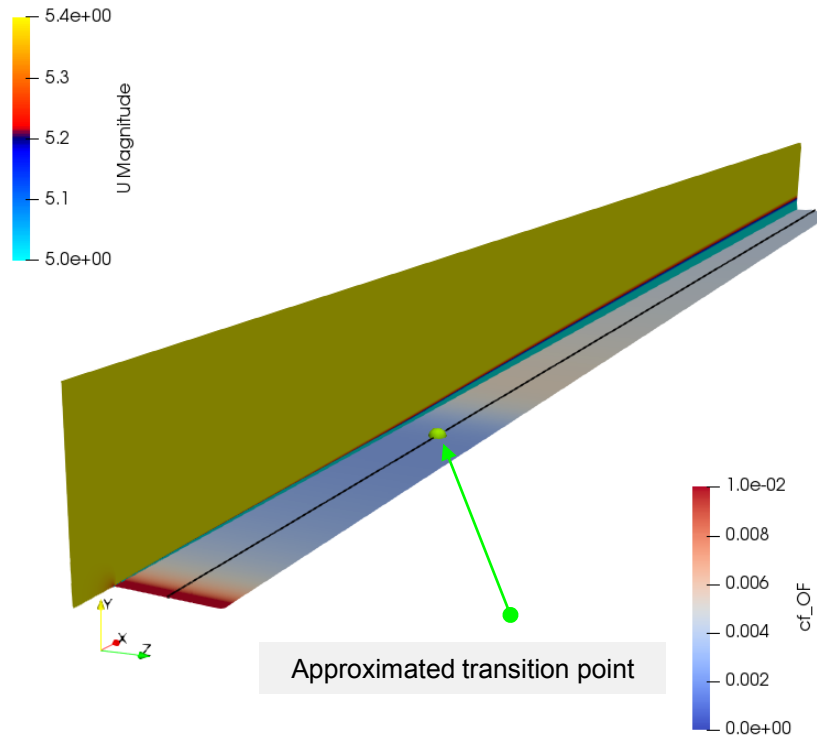
- **Guided tutorial 3.** Transitional flat plate.
- This case is ready to run.
- The case is located in the directory:

\$TM/turbulence/GT3/

- In the case directory, you will find a few scripts with the extension `.sh`, namely, `run_all.sh`, `run_mesh.sh`, `run_sampling.sh`, `run_solver.sh`, and so on.
- These scripts can be used to run the case automatically by typing in the terminal, for example,
 - `$> sh run_solver`
- These scripts are human-readable, and we highly recommend you open them, get familiar with the steps, and type the commands in the terminal. In this way, you will get used with the command line interface and OpenFOAM commands.
- If you are already comfortable with OpenFOAM, run the cases automatically using these scripts.
- In the case directory, you will also find the `README.FIRST` file. In this file, you will find some additional comments.

Guided tutorials

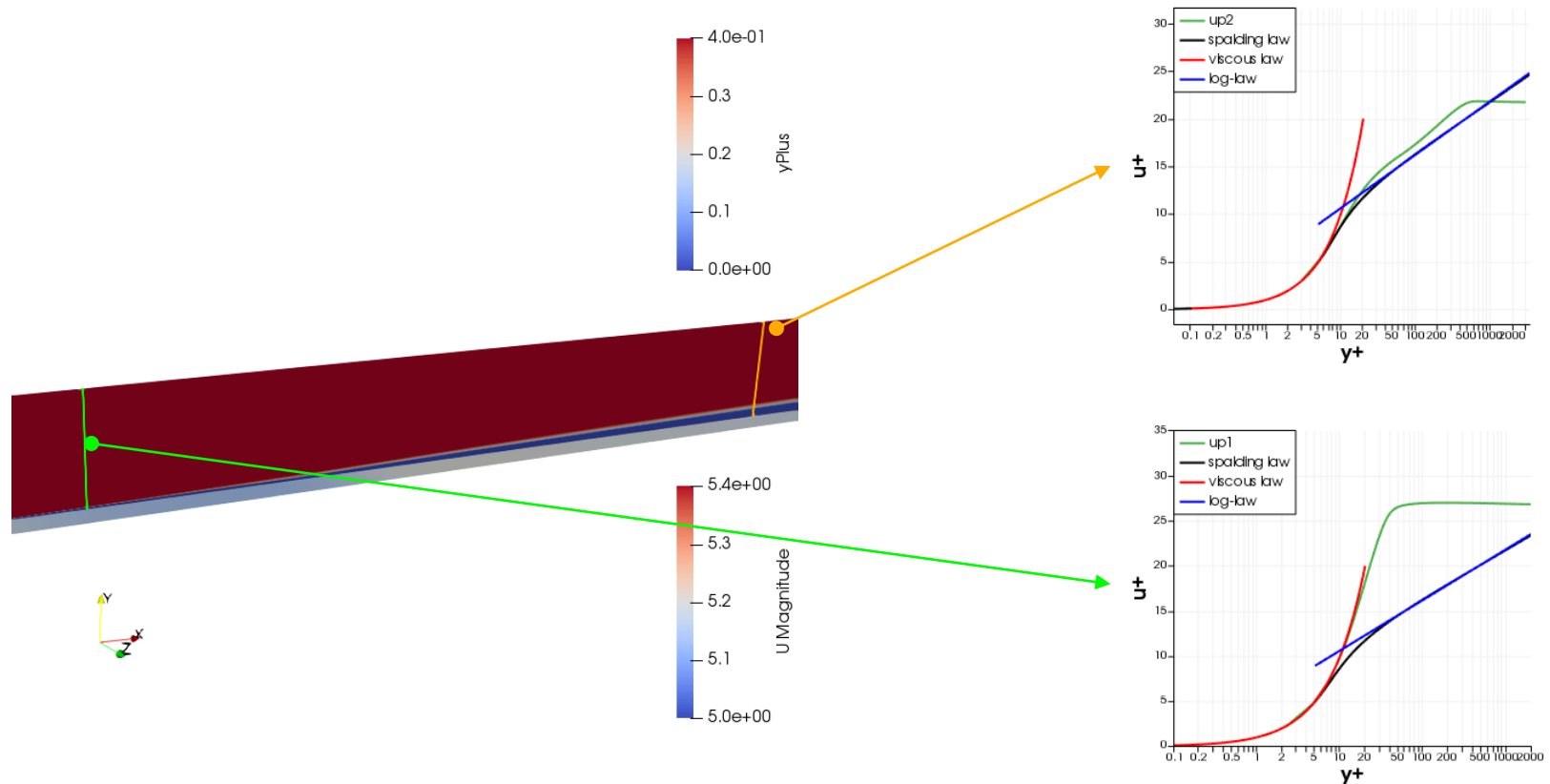
Guided tutorial 3 – Transitional flat plate



- In this guided tutorial we will simulate a transitional flow.
- We will use the classical 2D Zero Pressure Gradient Flat Plate validation case (ERCOFTAC T3A 3% test-case).
- Simulating transition to turbulence is very challenging and it requires very fine meshes and well calibrated methods.

Guided tutorials

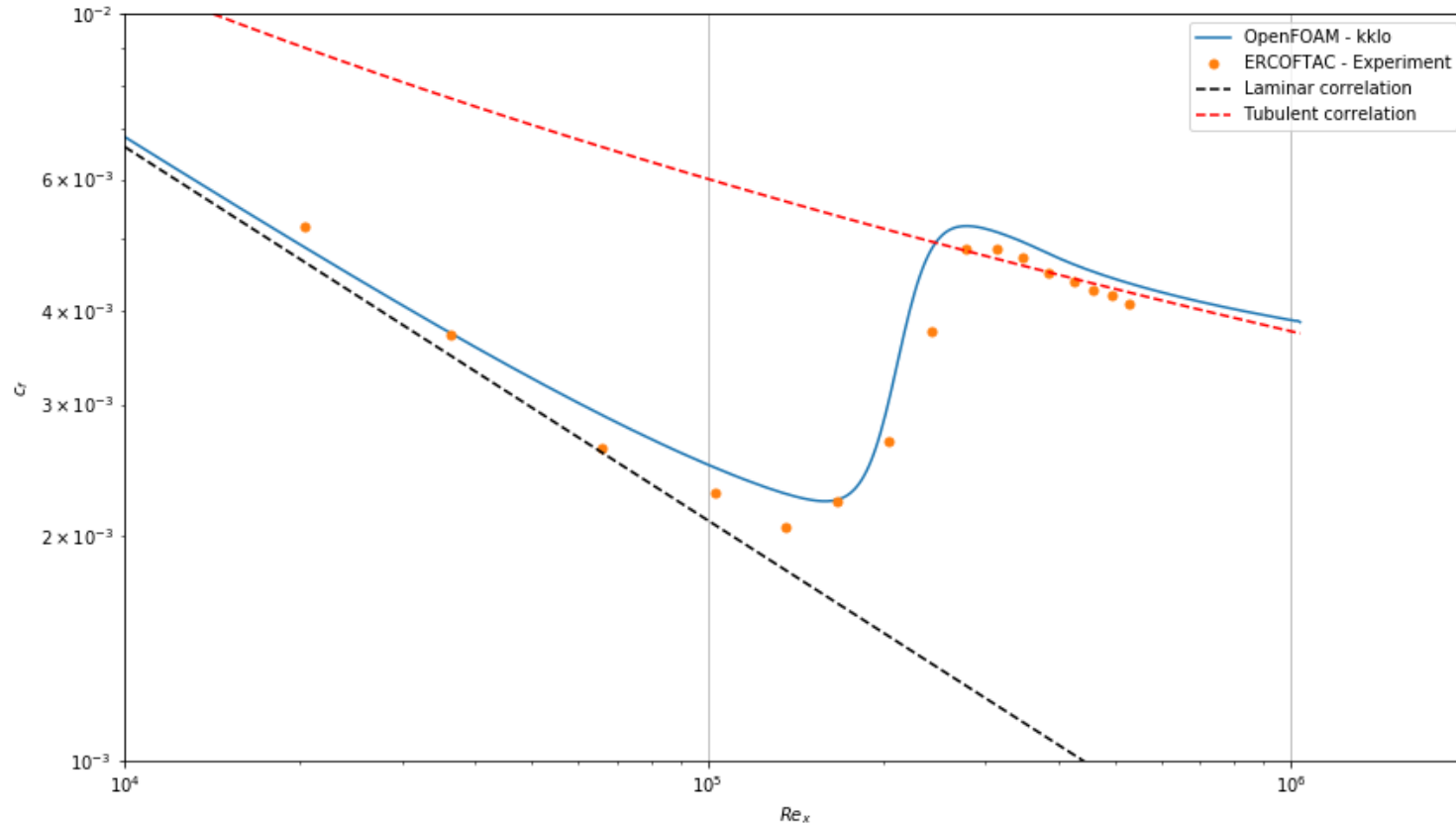
Guided tutorial 3 – Transitional flat plate



- In this guided tutorial we will simulate a transitional flow.
- We will use the classical 2D Zero Pressure Gradient Flat Plate validation case (ERCOFTAC T3A 3% test-case).
- Simulating transition to turbulence is very challenging and it requires very fine meshes and well calibrated methods.

Guided tutorials

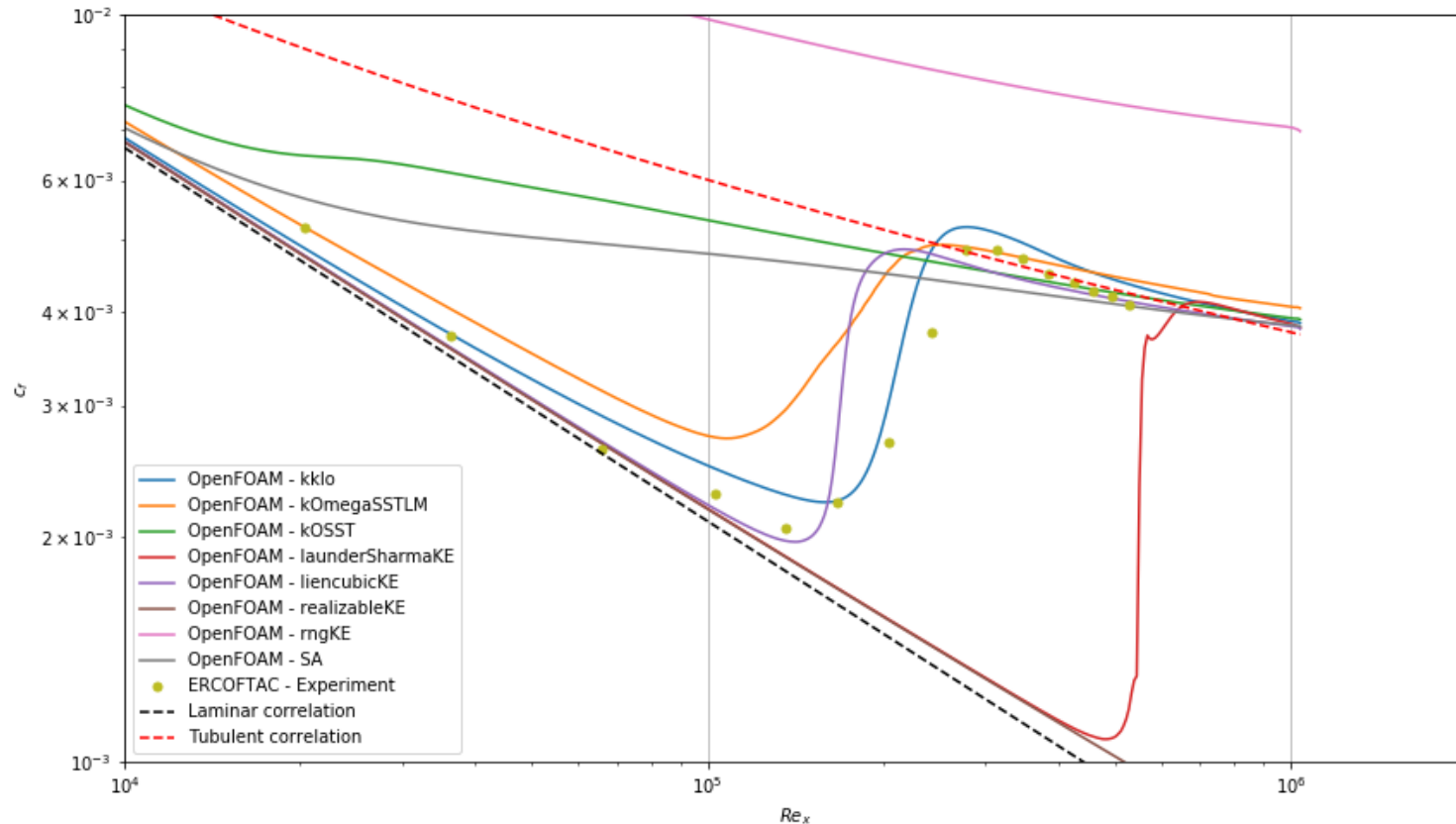
Guided tutorial 3 – Transitional flat plate



- Transition to turbulence – ERCOFTAC T3A 3% test-case.
- Maybe, this is the most challenging part in turbulence modelling as nobody knows what is happening in the transition region.

Guided tutorials

Guided tutorial 3 – Transitional flat plate



- Transition to turbulence – ERCOFTAC T3A 3% test-case.
- Maybe, this is the most challenging part in turbulence modelling as nobody knows what is happening in the transition region.

Guided tutorials

Guided tutorial 3 – Transitional flat plate

- When dealing with turbulence, setting boundary and initial conditions is of paramount importance.
- Our best advice is to get familiar with the theory.
- In the NASA Turbulence Modeling Resource site, you will find plenty of information:
<https://turbmodels.larc.nasa.gov/>
- Remember, transition to turbulence models are wall resolving.
- If you are using the $\gamma - Re_\theta$ model, the boundary and initial conditions for k and ω can be set as for the standard $k - \omega$ omega.
- The wall boundary and inlet/initial conditions for Re_θ are zero gradient and the values obtained from empirical correlations (using TU), respectively.
- The wall boundary and inlet/initial conditions for γ are zero gradient and 1, respectively.

Guided tutorials

Guided tutorial 3 – Transitional flat plate

- At the following links,

http://www.ercoftac.org/special_interest_groups/15_turbulence_modelling/sig_15_test_case_database/

http://cfd.mace.manchester.ac.uk/cgi-bin/cfddb/prpage.cgi?20&EXP&&database/cases/case20&cas20_head.html&cas20_desc.html&cas20_meth.html&cas20_data.html&cas20_refs.html&cas20_rsol.html&0&1&0&1&1&unknown

You will find more information about the ERCOFTAC T3A 3% transition test case.

Guided tutorials

Guided tutorial 3 – Transitional flat plate

- We are going to use the `incompressibleFluid` modular solver in steady (RANS) and unsteady (LES) modes.
- This case is rather simple, but we will use it to explain many features used in OpenFOAM when dealing with transition to turbulence and fully turbulent flows.
- When dealing with transitional models, we do not use wall functions.
- We will also show how to do the post-processing in order to reproduce the law of the wall. For this, we will use a jupyter notebook (or a python script) and paraview. Both approaches are valid.
- Remember, as we are introducing new closure equations for the turbulence problem, we need to define initial and boundary conditions for the new variables.
- We also need to define the discretization schemes and linear solvers to use to solve the new variables.
- It is also a good idea to setup a few functionObjects, such as: `y+`, minimum and maximum values, forces, time average, and online sampling.
- You will find the instructions of how to run this case in the file `README.FIRST` located in the case directory.